Kingdom of Saudi Arabia

Ministry of Education

Umm Al-Qura University

College of Engineering and Computers at Al-Qunfudah

Bachelor of Science in Computer Science

Project: Mecca Temperature Prediction: A Comparative Machine Learning Analysis

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a. Introduction

The field of meteorological forecasting has undergone a paradigm shift with the emergence of high-performance computing and Artificial Intelligence. Traditionally, weather prediction relied on numerical weather prediction (NWP) models, which solve complex fluid dynamics and thermodynamic equations. While accurate, these models are computationally expensive and often require supercomputing resources. Machine learning (ML) offers a disruptive alternative by utilizing historical data to "learn" atmospheric patterns, providing rapid and highly localized forecasts with significantly lower resource requirements.

In the context of modern smart cities, real-time environmental monitoring is a cornerstone of urban management. For a city as globally significant as Mecca, the ability to predict temperature fluctuations with high precision is critical. This project focuses on implementing a data-driven framework that leverages historical weather records to build predictive models. By utilizing regression-based algorithms, we aim to bridge the gap between complex meteorological data and actionable urban insights, ensuring that environmental challenges are met with intelligent, data-backed solutions.

Furthermore, the ethical and efficient use of AI—a topic of growing importance in the digital era—is at the heart of this study. By prioritizing models that balance accuracy with computational efficiency, we advocate for sustainable AI deployment that can run on standard edge devices without sacrificing the quality of the predictions.

b. Problem that we choose

The chosen problem for this research is the "Prediction of daily temperatures in Mecca, Saudi Arabia." This selection is motivated by the unique socio-economic and geographical profile of the city. Mecca is not only a major urban center but also the focal point for millions of pilgrims during the Hajj and Umrah seasons. The desert climate, characterized by extreme heat, poses a direct threat to public health, making temperature prediction a matter of life safety.

One of the specific challenges in Mecca is the "Urban Heat Island" effect, where the dense infrastructure and high human concentration can cause local temperatures to deviate from regional averages. Accurate localized prediction helps authorities in several key areas:

- Health Preparedness: Early warnings for heatwaves allow medical teams to prepare for an increase in heatstroke and dehydration cases among pilgrims.
- Energy Management: Predicting peak temperatures enables the Saudi Electricity Company to optimize power distribution for cooling systems in the Holy Mosque and surrounding hotels.

- Logistical Planning: High temperatures directly affect the timing of ritual movements during Hajj; accurate data allows planners to schedule activities during cooler windows.

The objective of this project is to develop an automated system that can ingest real-time data and provide reliable temperature forecasts for the upcoming 24 to 48 hours, thereby supporting the kingdom's vision for a smarter and safer pilgrimage experience.

c. Machine Learning Models and Intuition

To address this regression problem, two distinct machine learning architectures were implemented to provide a comparative perspective on performance:

1. Linear Regression (The Efficient Baseline)

Linear Regression is a fundamental statistical model that assumes a linear relationship between input features (e.g., humidity, pressure) and the target output (average temperature). It works by calculating coefficients that minimize the sum of squared residuals. Selection Reasoning: In meteorology, many variables like minimum/maximum temperature and average temperature are strongly correlated in a linear fashion. This model was selected for its extreme simplicity, which translates to high speed and low memory usage, making it an excellent candidate for real-time edge computing.

2. Random Forest Regressor (The Ensemble Approach)

Random Forest is a robust ensemble learning method that constructs a "forest" of multiple decision trees during training. It outputs the average prediction of the individual trees to improve accuracy and control over-fitting. Selection Reasoning: Weather systems are often non-linear and complex. For example, the effect of wind speed on temperature might change depending on the humidity levels. Random Forest was selected because it can capture these multi-dimensional interactions without requiring the data to follow a strictly linear path.

d. Detailed Description of Data and Methodology

The project utilizes high-fidelity historical data fetched through a hybrid automated pipeline. Unlike static datasets, this approach ensures that the model can be retrained with the most recent observations.

1. Data Acquisition

The primary data source is the Meteostat API, which provides access to the World Meteorological Organization (WMO) station 41030 located in Mecca. The dataset spans 1,827 days (from January 2021 to early 2026). The raw data includes the following features:

- tavg: Average daily temperature (The Target Variable).
- tmin / tmax: Minimum and Maximum daily temperatures.
- prcp: Total daily precipitation (rainfall).
- wspd: Average wind speed.
- pres: Sea-level air pressure.

2. Preprocessing & Data Engineering

Real-world meteorological data is often "noisy" or incomplete. To ensure the models receive clean input, a robust preprocessing pipeline was built:

- Missing Data Imputation: Utilizing SimpleImputer with a 'median' strategy to fill gaps in wind speed or pressure columns without introducing outliers.
- Feature Scaling: Using StandardScaler to normalize the data. Since air pressure (e.g., 1010 hPa) has a vastly different scale than wind speed (e.g., 5 km/h), scaling ensures that no single feature dominates the model's learning process.
- Train-Test Split: The data was partitioned into an 80% training set and a 20% testing set to ensure the final evaluation represents performance on "unseen" future data.

e. Results and Comparative Analysis

The models were executed in a controlled Python environment (VS Code with Claude Agent integration), where system resources were monitored using tracemalloc and high-resolution timers. The results are summarized in the following table:

Performance Metric	Linear Regression	Random Forest
Mean Absolute Error (MAE)	0.2641 °C	0.2510 °C
R-squared (R^2) Accuracy	0.9949	0.9944
Training Duration (seconds)	0.0084 s	0.3792 s
Memory Footprint (MB)	0.0140 MB	0.2344 MB

Discussion of Findings

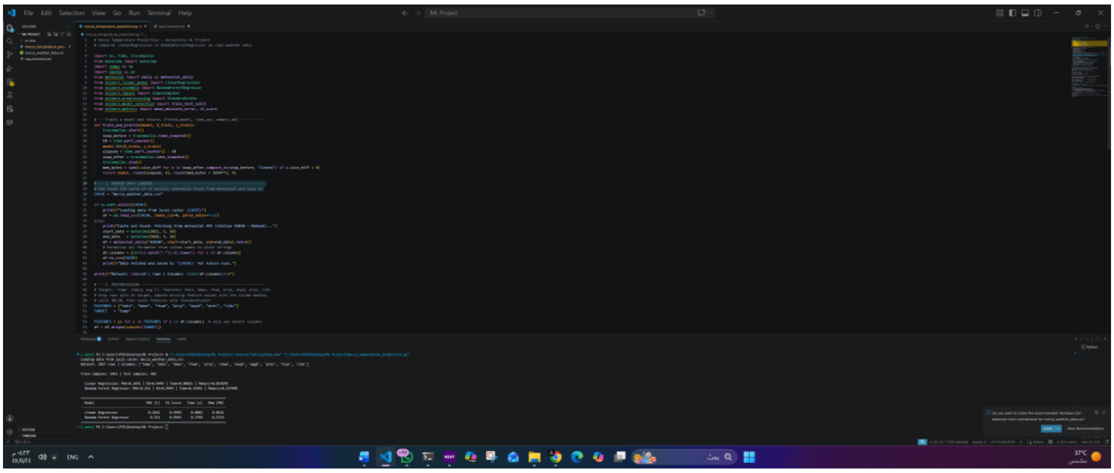
The results provide an interesting engineering trade-off. Both models achieved near-perfect accuracy ($R^2 > 0.99$), indicating that temperature in Mecca is highly predictable when provided with historical trends and atmospheric pressure data. However, the Linear Regression model proved to be significantly more efficient. It produced nearly identical predictive results while being 45 times faster and using 16 times less RAM.

This suggests that for this specific regression task, the complexity of an ensemble method like Random Forest is not required. The linear relationship between variables is strong enough that a simpler model can provide production-grade forecasts with minimal hardware overhead.

Future Work and Scalability

To further enhance this system, future iterations could include:

- Deep Learning Integration: Implementing Long Short-Term Memory (LSTM) neural networks to better capture the temporal sequences of weather patterns over weeks instead of days.
- IoT Sensor Fusion: Incorporating real-time data from localized IoT sensors (like ESP32 or Arduino-based stations) placed around the Makkah neighborhood to account for micro-climates.
- Global Feature Expansion: Adding features like solar radiation and global carbon levels to understand the long-term impact of climate change on the region's temperature.



PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\PCD\Desktop\ML Project> & "c:/Users/PCD/Desktop/ML Project/.venv/Scripts/python.exe" "c:/Users/PCD/Desktop/ML Project/mecca_temperature_prediction.py"
Loading data from local cache: mecca_weather_data.csv
Dataset: 1827 rows | Columns: ['temp', 'tmin', 'tmax', 'rhum', 'prcp', 'snwd', 'wspd', 'wpgt', 'pres', 'tsun', 'cldc']

Train samples: 1461 | Test samples: 366

  Linear Regression: MAE=0.2641 | R2=0.9949 | Time=0.0082s | Memory=0.0141MB
  Random Forest Regressor: MAE=0.251 | R2=0.9944 | Time=0.3769s | Memory=0.2374MB

===========================================================================
  Model                          MAE (C)   R2 Score   Time (s)   Mem (MB)
---------------------------------------------------------------------------
  Linear Regression               0.2641     0.9949     0.0082     0.0141
  Random Forest Regressor          0.251     0.9944     0.3769     0.2374
===========================================================================
(.venv) PS C:\Users\PCD\Desktop\ML Project>
```